MILESTONE 2 · DATASET COLLECTION & PREPROCESSING

Hindi → Haryanvi Text-to-Speech

Building the data foundation for a two-stage speech synthesis pipeline

NLP | Low-Resource Dialect | Bangru / Haryanvi | TTS | LLM Fine-tuning

TEAM

Contributions

PREPROCESSING & FINE-TUNING

Abhishek

- Conducted all further preprocessing
- Managed LLM fine-tuning tasks: dataset formatting (instruction-response), token length considerations, and prompt structuring

AUDIO & DATA UPLOAD

Satyam

- Created structured folders for audio/text datasets and uploaded to Hugging Face
- Extracted 3 audio files from YouTube using `yt-dlp`
- Scraped Haryanvi-Hindi parallel dataset from Storyweaver

SYNTHETIC DATA

Sanket

- Developed Python script for synthetic text data-generation and augmentation
- Documented report on synthetic data-generation

TEXT DATA & REPORT

Fazlur

- Scraped text data from `google-research-datasets/indic-gen-bench`
- Authored Milestone-2 report

PROJECT OVERVIEW

Two-Stage Pipeline

INPUT
Hindi Text
Standard written Hindi

→

STAGE 1
Text → Text
Hindi → Haryanvi (Bangru) via LLaMA 3.1

→

STAGE 2
Text → Speech
Haryanvi TTS via Sarvam / VITS

→

OUTPUT
Haryanvi Audio
Natural spoken Haryanvi

MODULE A

Text Dataset

Parallel Hindi–Haryanvi sentence pairs for translation model fine-tuning

MODULE B

Audio Dataset

Haryanvi audio + transcripts for TTS model fine-tuning

DATASET OVERVIEW

Two Independent Pipelines

ASPECT	MODULE A — TEXT	MODULE B — AUDIO
Task	Sequence-to-sequence translation	Text-to-speech synthesis
Input	Hindi sentences	Haryanvi text transcripts
Output	Haryanvi sentences	Haryanvi audio waveforms
Primary Format	CSV / JSON	Audio array + transcript
Dataset Size	5,594 pairs	5,515 clips (~1.89 GB)
Target Split	80 / 10 / 10 (train/val/test)	90 / 10 (train/eval)

A

MODULE A

Text Dataset
Hindi → Haryanvi Translation

Text Corpus Sources

SOURCE 1 · PRIMARY

XOR-QA Indic (Google Research)

- 594 English–Haryanvi QA pairs
- English → Hindi via Google Translate
- Manual spot-check applied
- Open research license (Google open-data policies)
- URL: `indic-gen-bench / xorqa_bgc_dev.json`

SOURCE 2 · PRIMARY

Synthetic Data (Google Gemini)

- 5,000 parallel Hindi–Haryanvi pairs
- Research-grade prompt enforcing Bangru rules
- Diverse topics & sentence lengths (16 domains)
- Labelled as LLM-generated; segregated in eval
- Format: CSV + JSON (`synthetic_haryanvi_corpus.json`)

FALLBACK

Source 3 — Storyweaver (Pratham Books)

Not currently used. Haryanvi-Hindi parallel corpus scraping. Open-source; licensed under Creative Commons with attribution required. Reserved if primary corpus proves insufficient for model convergence.

Bangru Linguistic Rules Enforced

Morpho-syntactic Substitutions

है	→	सै
हूँ	→	सूँ
हैं	→	से
रहा/रही/रहे	→	रया/रयी/रये
नहीं	→	कोनी
ने	→	नै

Lexical Replacements

मुझे/मैंने	→	मन्ने
तुम्हें/तुमने	→	तन्ने
हमारा / तुम्हारा	→	म्हारा / थारा
यह / वह	→	यो / वो
बहुत / कहाँ / कब	→	घणा / कड़े / कदे

Topic Coverage: Daily life, farming, rural culture, family, markets, school, travel, food, sports, festivals, weather, technology, government services, and transportation — ensuring broad domain generalization.

Quality Controls

QUALITY CONSTRAINTS APPLIED

- No sentence duplication
- No literal word-by-word translation
- No untranslated Hindi words where Haryanvi equivalent exists
- Varied sentence complexity to prevent **dataset collapse**
- 16 topic domains covered for broad domain generalization

Text Dataset Cleaning Pipeline

STEP 1

Missing Value Check

Column-wise null check via `df.isnull().sum()`. Rows with null Hindi or Haryanvi entries dropped.

STEP 2

Deduplication

Exact duplicates removed via `drop_duplicates()` before splitting. Count logged for traceability.

STEP 3

Length Filtering

Hindi sentences: min **3 words**, max **100 words**. Removes fragments and extreme outliers.

STEP 4

Manual Spot-check

Subset of XOR-QA pairs reviewed for Haryanvi authenticity. Google Translate artifacts flagged and corrected.

SOURCE	RAW SAMPLES	RETAINED AFTER CLEANING
XOR-QA Indic (GitHub)	594	594
Synthetic (Gemini-generated)	5,000	5,000
Total	**5,594**	**5,594**

Dataset Splits & LLM Formatting

Train / Val / Test Split

Train 80% (~4,475) Val 10% (~560) Test 10% (~560)

Leakage Prevention: Deduplication before split. Fixed seed `42` via `sklearn train_test_split`. No temporal or source-based leakage.

LLaMA 3.1 Chat Template Format

- **System:** Hindi→Haryanvi translator persona with morpho-syntactic rules
- **User turn:** Hindi source sentence
- **Assistant turn:** Target Haryanvi sentence
- **Max length:** 128 tokens, padded, truncated
- **Output:** HuggingFace `DatasetDict` with train/validation/test splits

B

MODULE B

Audio Dataset Haryanvi TTS

Audio Corpus Sources

SOURCE 1 · PRIMARY ACTIVE

HuggingFace — ankitdhiman/haryanvi-tts

5,515
clips

- 5,515 audio–text pairs in Bangru dialect
- ~1.89 GB · AudioFolder format · `metadata.csv`
- High quality, pre-aligned, research-ready
- No explicit license — confirm with author before commercial use
- Loaded via `load_dataset("ankitdhiman/haryanvi-tts")`

SOURCE 2 · FALLBACK NOT ACTIVE

YouTube Audio Extraction (Haryanvi Channels)

Activated only if HuggingFace data falls below ~5–10 hour threshold. Pipeline: `yt-dlp` + FFmpeg → MP3 192kbps. Transcription via `youtube-transcript-api` or Whisper tiny fallback.

3 videos identified · Academic use only · YouTube ToS compliance required

MODULE B · FALLBACK SOURCES

Identified YouTube Videos

#	URL
Video 1	https://www.youtube.com/watch?v=Re7OOS6wD0s
Video 2	https://www.youtube.com/watch?v=4dtm-FVDBEs
Video 3	https://www.youtube.com/watch?v=D614BjIiRzU

If activated: Convert MP3 → WAV at 22,050 Hz mono · Segment by Whisper timestamps · Manual verification required · Speaker/video-stratified splitting to prevent leakage · Log video ID per clip.

MODULE B · QUALITY STANDARDS

Audio Quality Parameters

22,050
Hz target sample rate

1.0s
minimum clip duration

15.0s
maximum clip duration

90/10
train / eval split

RESAMPLING

Programmatic via HuggingFace

All audio cast using `Audio(sampling_rate=22050)`. Consistent regardless of original source format. Standard for VITS-based TTS models.

OUTPUT FORMAT

TTS-Ready DatasetDict

`audio_array` — NumPy waveform at 22,050 Hz

`normalized_text` — Whitespace-stripped Haryanvi transcript

MODULE B · FALLBACK PIPELINE

YouTube Extraction — If Activated

1
yt-dlp
Download best audio → MP3 192kbps

→

2
Transcript
YT API first, Whisper tiny as fallback

→

3
Convert
MP3 → WAV 22,050 Hz mono

→

4
Segment
Clip by Whisper timestamps · 1–10s

→

5
Ingest
Into rds.py with video-ID tag

Note: Whisper `tiny` used for speed. Upgrade to `medium` or `large-v3` recommended if fallback is activated for production data collection. Speaker/video-stratified splitting required to prevent train/eval leakage.

ADEQUACY ASSESSMENT

Is the Data Sufficient?

MODULE A — TEXT

5,594 Parallel Pairs

- Adequate starting point for LLM fine-tuning on a low-resource dialect task
- Synthetic data justified — extreme scarcity of Hindi–Haryanvi parallel corpora
- Bangru morphology enforced in prompt → reduces dialect drift risk
- **If insufficient:** Storyweaver scraping activated as next step

MODULE B — AUDIO

5,515 Clips · ~1.89 GB

- High-quality, pre-aligned baseline from HuggingFace
- Total usable hours to be confirmed after 1–15s filtering
- **If < 5–10 hrs:** Activate YouTube fallback
- **Or:** Transfer-learn from Hindi TTS (IndicTTS / Sarvam-TTS)

INTER-TASK ALIGNMENT

Coupling Module A ↔ Module B

MODULE A OUTPUT
Haryanvi Text
Devanagari script

→

CRITICAL LINK
Text Normalization
Shared function — both modules

→

MODULE B INPUT
TTS Transcript
Must match training format

SCRIPT CONSISTENCY

Devanagari Throughout

Vocabulary and script in text training data must match transcript format expected by TTS model — consistent Haryanvi spellings required across both modules.

⚠ OPEN ITEM

Shared Normalization Function

A unified text normalization function (anusvara, nukta, punctuation handling) to be implemented before inference. Silent failures otherwise.

ETHICAL CONSIDERATIONS

Data Sourcing & Usage

SOURCE	STATUS	CONSTRAINTS
XOR-QA (Google Research)	✓ Open	Academic use; Google open-data policy
Storyweaver (Pratham Books)	✓ CC Licensed	Creative Commons; attribution required; fallback only
Synthetic (Gemini)	✓ Labelled	Clearly LLM-generated; segregated in eval
HuggingFace TTS	⚠ Unlicensed	Research only; confirm with author before commercial use
YouTube Audio	⚠ Fallback	Academic fair-use; YouTube ToS compliance required
Representation	✓ Diverse	16 topic domains; avoids cultural stereotyping

REPRODUCIBILITY

Checklist

- ✓ Fixed random seed (`RANDOM_SEED = 42`) throughout
- ✓ All thresholds in central `DataConfig` class
- ✓ Duplicate removal logged before splitting
- ✓ HuggingFace `DatasetDict` for consistent loading
- ✓ Audio resampled programmatically to 22,050 Hz
- ✓ Per-source counts documented (XOR-QA: 594, Synthetic: 5,000)
- ✓ YouTube pipeline documented; video URLs logged
- ○ Total audio hours — pending duration filter run
- ○ Shared text normalization function — pending implementation

Implemented across two Colab notebooks: `rds.py` (main pipeline) · `youtube_video_to_audio.py` (fallback extraction)

REQUIREMENTS

Key Dataset Requirements for Training

MODULE A — TRANSLATION MODEL

- Parallel Hindi–Haryanvi pairs with authentic Bangru morphology
- Diverse sentence lengths (5 words to multi-sentence) and domains
- Grammatically correct Haryanvi per documented substitution rules
- No leakage between train/val/test; duplicates removed before splitting
- Current corpus: **5,594 pairs**

MODULE B — TTS MODEL

- Audio at **22,050 Hz** mono (VITS/Sarvam-TTS standard)
- Clip duration between **1.0–15.0 seconds**
- Accurate text-audio alignment
- Speaker diversity across clips for voice generalization
- Primary: `ankitdhiman/haryanvi-tts` (5,515 clips); YouTube fallback available

SUMMARY

Milestone 2 — Key Takeaways

TEXT PIPELINE

- **5,594 Hindi–Haryanvi pairs** from 2 active sources
- Strict Bangru morphological rules in synthetic generation
- 80/10/10 split · no leakage · seed 42
- Formatted for LLaMA 3.1 instruction fine-tuning

AUDIO PIPELINE

- **5,515 clips (~1.89 GB)** from HuggingFace
- 22,050 Hz · 1–15s duration window
- 90/10 train/eval split · seed 42
- YouTube fallback pipeline (3 videos) ready if needed

Data is ethically sourced, quality-assessed, and structured for direct ingestion into Stage 1 (LLaMA translation) and Stage 2 (Sarvam/VITS TTS) fine-tuning pipelines.